KARTHIKEYAN S

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Education

Sri Krishna College of Engineering and Technology **2023 – 2027**
B.Tech in Artificial Intelligence and Data Science, CGPA: 8.63/10 *Coimbatore, Tamil Nadu, India*

Technical Skills

Languages: Java, C++, Python

Backend & Frameworks: Spring Boot, REST APIs, React.js

Databases & Cloud: MySQL, Docker, AWS

Tools: Git, GitHub, Postman, Linux

Core CS: Data Structures & Algorithms, Operating Systems, System Design (LLD), GenAI

Experience

Java Programming Intern **Feb 2025 – April 2025**
Infosys Springboard *Virtual*

- Learned **Spring Boot** fundamentals, including project structure, controllers, services, and dependency injection, to build Java backend applications.
- Learned to build and test **RESTful APIs** using HTTP methods and **JSON** with Postman, and picked up **React.js** basics (components, props, state, Hooks) to see how a frontend consumes APIs.
- Practiced basic database operations (CRUD) with **MySQL** and **Spring Data JPA**, connecting the backend to a relational database.

Projects

AgeVision – Facial Age Estimation using Deep Learning *Python · TensorFlow · ResNet50 · OpenCV · Transfer Learning*

- Built a face detection and alignment preprocessing pipeline with automated quality gating (Laplacian-variance blur rejection, lighting/pose checks) to filter low-quality input before inference.
- Fine-tuned a ResNet50 backbone via transfer learning with a regression head to predict age, optimizing Mean Absolute Error (MAE) on a 185,000+ image dataset (ages 1–100) rebalanced across age groups to address class imbalance.

Explainable AI Decision-Support System for Diabetic Retinopathy *Python · TensorFlow · EfficientNetB3 · OpenCV · Grad-CAM*

- Fine-tuned an EfficientNetB3 backbone for 5-class ICDR severity grading of fundus images, backed by automated Laplacian/luminance-based image-quality screening algorithms.
- Engineered Grad-CAM interpretability and TTA-based uncertainty estimation, plus an OLS regression module forecasting patient disease trajectory over time – translating an ambiguous, open-ended clinical problem into a structured, quantifiable solution.

Multicloud File Storage Application *Java · Spring Boot · REST API · Google Drive API · OneDrive API*

- Designed a provider-agnostic Strategy Pattern abstraction unifying Google Drive's graph model and OneDrive's path hierarchy under a single scalable **API contract**, applying object-oriented design to decouple provider-specific logic.
- Implemented streaming file transfers with AES-256 encrypted token storage, preventing OutOfMemoryErrors on large uploads and improving fault tolerance for large-scale file operations.

Certifications & Achievements

- **AWS Cloud Practitioner** – Amazon Web Services
- **Java Foundation Certificate** – Infosys Springboard
- **Spring Data JPA with Spring Boot** – Infosys Springboard
- **AI Foundation Certification** – Infosys Springboard
- **LeetCode:** 570+ problems solved, primarily in Java
- **LeetCode Contests:** Rating of 1700+ (weekly & biweekly)